Chapter 5 Breakout (Pt. II)

STAT 4714

Questions

$n = 6$

1. A homeowner plants 6 flower bulbs selected at random from a box containing 5 tulip bulbs and 4 daffodil bulbs. What is the probability that they planted 2 daffodil bulbs and 4 tulip bulbs?

$N = 5 + 4 = 9$

Hypergeometric Experiment:

- Choose 6 of 9 w/o replacement
- 2 classes: tulips & daffodils

X = # of tulips, w/ $k = 5$ total tulips to draw

$\Rightarrow h(x; N = 9, n = 6, k = 5)$

$P(X = 4) = h(4; N = 9, n = 6, k = 5)$

math

$$\frac{\binom{5}{4}\binom{4}{2}}{\binom{9}{6}} = 5/14$$

code

dhyper(4, 5, 9-5, 6)

$= 0.3571$

$n=7$

2. If 7 cards are dealt from an ordinary deck of 52 playing cards, what is the probability that

$N=52$

   (a) exactly 2 of them will be face cards?

   (b) at least 1 of them will be a queen?

(a) $X = \#$ of face ; $k = 4 \cdot 3 = 12$ face cards

$h(x; N=52, n=7, k=12)$

$P(X=2) = \text{dhyper}(2, 12, 40, 7) = \underline{0.3246}$

(b) $X = \#$ of queens; $k = 4$ queen cards

$h(x; N=52, n=7, k=4)$

$P(X \geq 1) = 1 - P(X=0)$

$$= 1 - \frac{\binom{4}{0}\binom{48}{7}}{\binom{52}{7}} = \underline{0.4496}$$

$N = 3 + 2 + 4 = 9$

$n = 5$

3. An urn contains 3 green balls, 2 blue balls, and 4 red balls. In a random sample of 5 balls, find the probability that both blue balls and at least 1 red ball are selected.

3 outcomes $\Rightarrow$ multivariate hypergeometric!

$X_1 = \#$ of green, $X_2 = \#$ of blue, $X_3 = \#$ of red

$$mh(x_1, x_2, x_3; N = 9, n = 5, a_1 = 3, a_2 = 2, a_3 = 4)$$

$$P(X_2 = 2, X_3 \geq 1)$$

$$= P(X_2 = 2, X_3 = 1, X_1 = 2)$$

$$+ P(X_2 = 2, X_3 = 2, X_1 = 1)$$

$$+ P(X_2 = 2, X_3 = 3, X_1 = 0)$$

$$= mh(2, 2, 1; N = 9, n = 5, 3, 2, 4)$$

$$+ mh(1, 2, 2; N = 9, n = 5, 3, 2, 4)$$

$$+ mh(0, 2, 3; N = 9, n = 5, 3, 2, 4)$$

or

$$= \frac{\binom{3}{2}\binom{2}{2}\binom{4}{1}}{\binom{9}{5}} + \frac{\binom{3}{1}\binom{2}{2}\binom{4}{2}}{\binom{9}{5}} + \frac{\binom{3}{0}\binom{2}{2}\binom{4}{3}}{\binom{9}{5}}$$

$$= 17/63$$

4. A government task force suspects that some manufacturing companies are in violation of federal pollution regulations with regard to dumping a certain type of product. Twenty firms are under suspicion but not all can be inspected. Suppose that 3 of the firms are in violation. $N = 20$

   (a) What is the probability that inspection of 5 firms will find no violations? $n = 5$

   (b) What is the probability that the plan above will find two violations?

2 outcomes = {violation, innocent}

$\Rightarrow$ hypergeometric

X = # of violations ; $k = 3$

$h(x; N = 20, n = 5, k = 3)$

(a) $P(X=0) = \dfrac{\binom{3}{0}\binom{17}{5}}{\binom{20}{5}} = 0.3991$

(b) $P(X=2) = \text{dhyper}(2, 3, 17, 5) = 0.1316$